Final Year Project Presentation

Identifying the Best Deployable Machine Learning Model for Malaria and Dengue Diagnosis

A Comparative Benchmarking and Deployment Framework for Low-Resource Healthcare Settings

University of South Asia

Department of Computer Science

2023 - 2027



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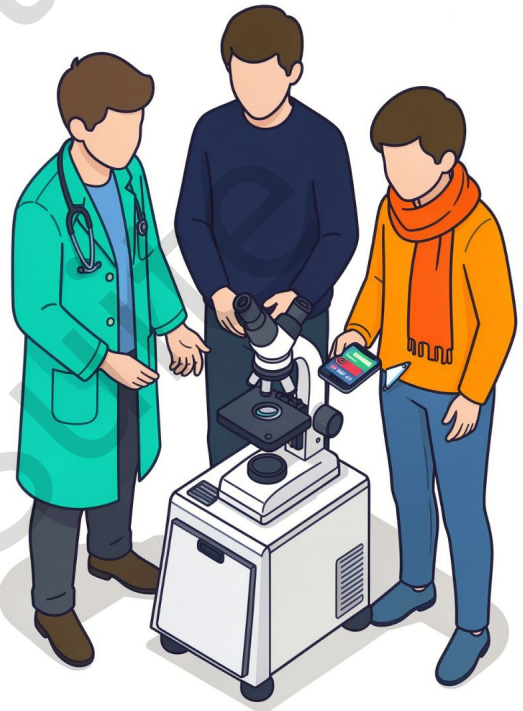
Project Team

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Introduction and Background

- Malaria and dengue are endemic in Pakistan and South Asia
- Both begin as undifferentiated fever
- **263 million** malaria cases and **~597,000** deaths (WHO, 2023)
- **14 million+** dengue cases (2024)
- Malaria microscopy depends on microscopist skill and workload
- Dengue tests (NS1/serology) are often unavailable in small facilities
- ML achieves high accuracy on smear images and routine blood values
- But studies use different splits and preprocessing, and rarely measure latency, memory, calibration, or explainability



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Problem Statement

- No standard protocol exists for choosing a deployable model in low-resource settings
- Literature is not comparable across studies
- Deployment constraints are ignored
- Result: model selection defaults to peak accuracy

Peak accuracy can favor models that are:

- Slow
- Large
- Overconfident
- Unexplainable

Accuracy alone is the wrong selection criterion.

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Existing Work: Reported Accuracy Benchmarks

Malaria: NIH cell-image dataset (27,558 images)

- Baseline CNN: ~92–96%
- MobileNetV2: ~93.5%
- ResNet-50 (transfer learning): ~96%
- Xception: >97%

Dengue: routine blood values (CBC)

- Random Forest: ~84.5% (1,523 patients, Bangladesh)
- XGBoost: ~81–87% (same dataset size, different papers)
- Random Forest: ~85% (Brazil)
- XGBoost / LightGBM: AUC >0.90 (49,679 patients, Puerto Rico)

Accuracy varies widely because every study uses a different dataset, split and preprocessing.

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Research Gaps

- Results are not comparable: different datasets, splits and preprocessing
- Accuracy is reported alone; latency, memory, calibration and explainability are rarely measured
- Clinical validation is limited
- Dengue models struggle on real febrile patients
- No method exists for choosing which model to deploy

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Proposed Solution and Novelty

Proposed Solution	Novelty
1 Build one lightweight custom model per disease	Deployability, not accuracy, as the selection criterion
2 Benchmark it against 2 established models per disease under one protocol	Clinical sensitivity gate (95% malaria, 85% dengue)
3 Score all models on diagnostic and deployment criteria with TOPSIS (Deployability Score)	Ranking-stability testing (weights varied ±10%)
4 Compress the top model per disease and demonstrate it in a prototype	Custom lightweight models designed for low-resource settings

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Datasets

Malaria: NIH Malaria Cell Images

- 27,558 segmented thin-smear cell images
- 13,779 parasitized + 13,779 uninfected
- From 150 infected and 50 healthy patients
- **Preprocessing:** resize to 224×224, normalize
- Split: 70/15/15, stratified

Infected



Not Infected



Dengue: Clinical CBC and Symptom Data

- Dataset: Dengue Fever Hematological Dataset
- Size: 1,523 patients, 13 features
- Classes: 1,040 dengue-positive and 481 dengue-negative
- Features: platelet count, WBC, haematocrit, neutrophils, lymphocytes, symptoms
- Preprocessing: duplicate removal, imputation, scaling, SMOTE within training folds only
- Validation: repeated stratified 5-fold cross-validation

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Candidate Models: 6 Models

Malaria: Cell Images

- **Custom CNN: built by us**
  (3–4 convolutional layers)
- **MobileNetV3:**
  established, designed for mobile
- **EfficientNet-B0:**
  established, accuracy/efficiency balance

Dengue: (blood values)

- **Custom MLP (Multi-Layer Perceptron)**
  built by us
- **Random Forest:**
  established, strong tabular baseline
- **XGBoost:**
  established, gradient-boosted trees

Why these models?
Small clinics need fast, small models that run offline on a phone or low-end computer.

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Project Objectives and Methodology

Objectives

1. Prepare malaria and dengue datasets under a documented protocol

2. Build one custom model per disease and train 2 established models for comparison

3. Measure diagnostic and deployment metrics for all 6 models

4. Rank with TOPSIS and a sensitivity gate, and test ranking stability

5. Quantify quantization and pruning trade-offs

6. Provide Grad-CAM and SHAP explanations with clinician ratings

7. Deploy the selected models in a minimal prototype

Methodology Pipeline

Data → Train → Compress → Score → Explain → Statistical Analysis

Metrices: sensitivity, specificity, precision, F1, ROC-AUC, calibration error, robustness, latency, memory, model size

Compression: INT8 quantization and pruning, exported to ONNX and TFLite

Explainability: Grad-CAM (images) and SHAP (blood values)

Tracking: all experiments logged in MLflow

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Requirements and System Architecture

Functional Requirements

- Enter lab values and symptoms, or upload a smear image
- Receive a prediction, confidence level, risk flag, and visual explanation
- Store all predictions for later review

Non-Functional Requirements

- Offline capability on Android
- Low latency and small model size
- Decision support only

MalDen-Deploy: Four-Layer Architecture

1. Presentation: web dashboard and Android app
2. Application: API (authentication, predictions, benchmarks, explanations)
3. AI/ML: optimised models, compression tools, scoring engine
4. Data: PostgreSQL, object storage, MLflow model registry

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Scope and Limitations

In Scope

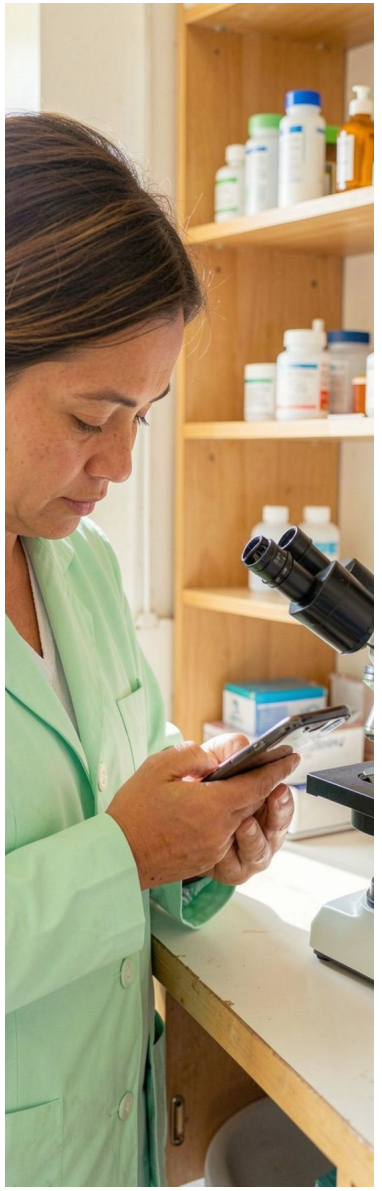
- Benchmarking for two tasks (malaria and dengue)
- Model selection and compression
- Explainability
- Prototype

Out of Scope

- Final diagnosis or treatment advice
- New laboratory hardware
- Clinical trials and regulatory approval
- Dengue serotyping and severe-dengue prediction

Risks and Mitigation

- Small dengue data → repeated cross-validation and external validation
- Quantisation accuracy loss → full-precision fallback
- Compute limits → university GPUs or Colab
- Ethics → anonymised data, ethics approval, decision support only



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Expected Outcomes and Deliverables

A repeatable benchmark of **6 models**

A data-backed recommendation of the best deployable model per disease

Compressed models with a measured size, speed, and accuracy trade-off

A working prototype: **MalDen-Deploy**

A framework reusable for other diseases

One custom model per disease, compared against established models

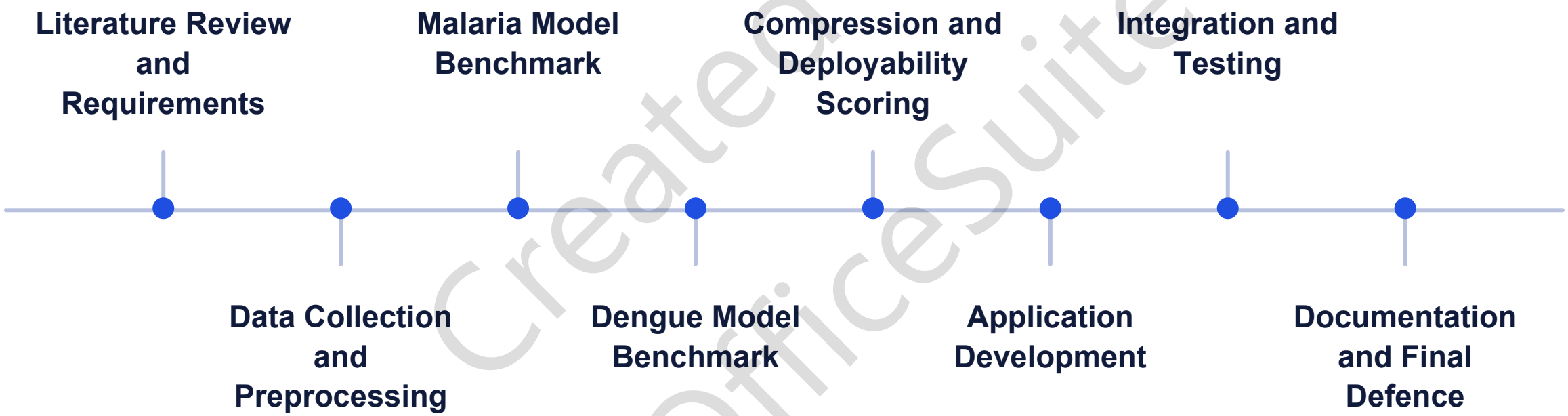


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Project Timeline and Work Plan

Approximately 8 months, with overlapping phases



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Conclusion, References and Q&A

Key Takeaway: The best model is not the most accurate. It is the most **deployable**.

References:

- WHO, World Malaria Report 2024
- Haider N. et al., Global dengue epidemic 2024, Int. J. Infect. Dis., 2025
- Rajaraman S. et al., PeerJ 6:e4568, 2018 (NIH malaria dataset)
- Vu D.M. et al., Dengue detection in Kenyan febrile children, PLOS Glob. Public Health, 2023
- Hwang & Yoon, Multiple Attribute Decision Making (TOPSIS), 1981

Dataset Links:

- Malaria Dataset (https://www.kaggle.com/datasets/iarunava/cell-images-for-detecting-malaria)
- Dengue Dataset (https://data.mendeley.com/datasets/6fsrsk3mb8)

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